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Image as a Foreign Language: BEIT Pretraining for All Vision and Vision-Language Tasks

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Abstract

A big convergence of language, vision, and multimodal pretraining is emerging. In this work, we introduce a general-purpose multimodal foundation model **BEIT-3**, which achieves state-of-the-art transfer performance on both vision and vision-language tasks. Specifically, we advance the big convergence from three aspects: backbone architecture, pretraining task, and model scaling up. We introduce Multi-way Transformers for general-purpose modeling, where the modular architecture enables both deep fusion and modality-specific encoding. Based on the shared backbone, we perform masked "language" modeling on images (**Imglish**), texts (English), and image-text pairs ("parallel sentences") in a unified manner. Experimental results show that BEIT-3 obtains state-of-the-art performance on object detection (COCO), semantic segmentation (ADE20K), image classification (ImageNet), visual reasoning (NLVR2), visual question answering (VQAv2), image captioning (COCO), and cross-modal retrieval (Flickr30K, COCO).

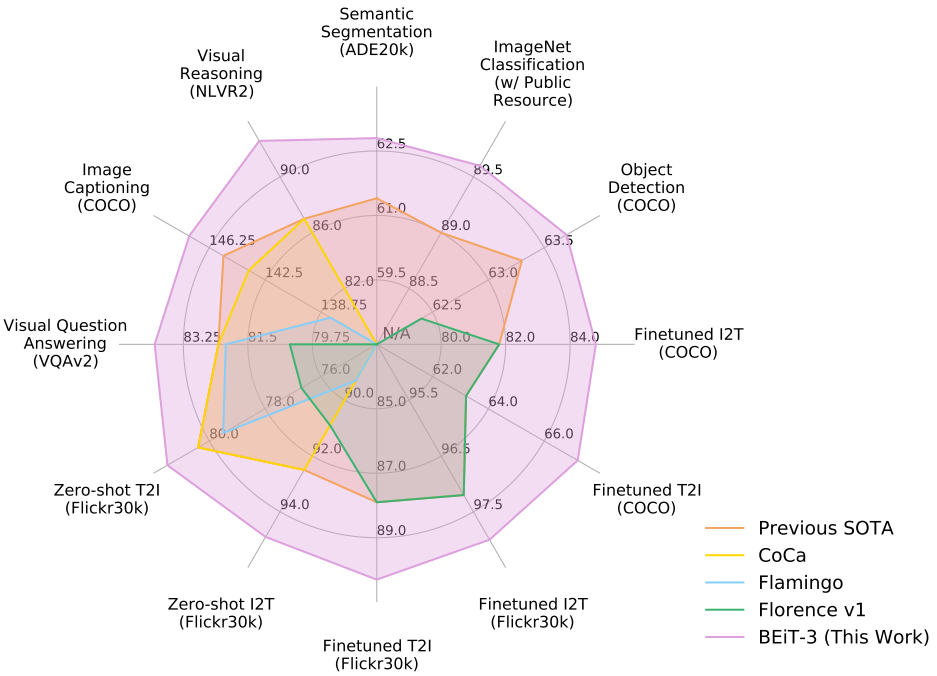


Figure 1: BEIT-3 achieves state-of-the-art performance on a broad range of tasks compared with other customized or foundation models. I2T/T2I is short for image-to-text/text-to-image retrieval.

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图像即外语：面向所有视觉和视觉-语言任务的 BEIT 预训练

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摘要

语言、视觉与多模态预训练正在出现大融合。在这项工作中，我们提出了通用多模态基础模型 BEIT-3，它在视觉任务和视觉-语言任务上都实现了最先进的迁移性能。具体而言，我们从三个方面推进了这种大融合：骨干架构、预训练任务以及模型规模扩展。我们引入了用于通用建模的 Multi-way Transformers，其模块化架构同时支持深度融合和特定模态编码。基于共享骨干，我们以统一方式对图像（Imglish）、文本（English）以及图像-文本对（“平行句子”）执行掩码“语言”建模。实验结果表明，BEIT-3 在目标检测（COCO）、语义分割（ADE20K）、图像分类（ImageNet）、视觉推理（NLVR2）、视觉问答（VQAv2）、图像描述生成（COCO）以及跨模态检索（Flickr30K、COCO）上取得了最先进的性能。

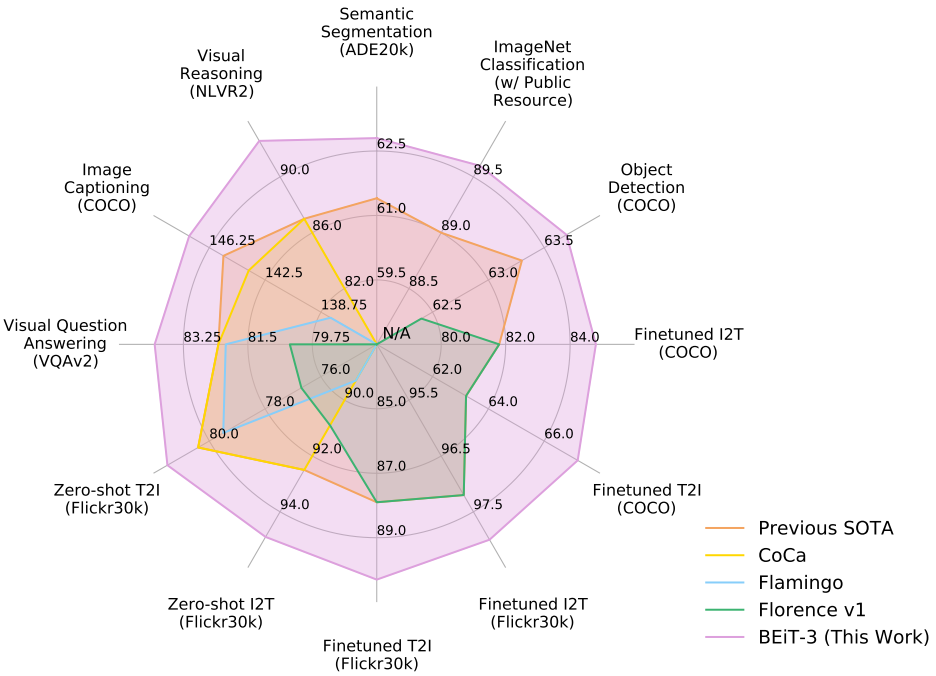


图 1：与其他定制模型或基础模型相比，BEIT-3 在广泛任务上实现了最先进的性能。I2T/T2I 是图文检索/文图检索的缩写。

* Equal contribution. † Corresponding author.

Category	Task	Dataset	Metric	Previous SOTA	BEIT-3
Vision	Semantic Segmentation	ADE20K	mIoU	61.4 (FD-SwinV2)	**62.8** (+1.4)
	Object Detection	COCO	AP	63.3 (DINO)	**63.7** (+0.4)
	Instance Segmentation	COCO	AP	54.7 (Mask DINO)	**54.8** (+0.1)
	Image Classification	ImageNet†	Top-1 acc.	89.0 (FD-CLIP)	**89.6** (+0.6)
Vision-Language	Visual Reasoning	NLVR2	Acc.	87.0 (CoCa)	**92.6** (+5.6)
	Visual QA	VQAv2	VQA acc.	82.3 (CoCa)	**84.0** (+1.7)
	Image Captioning	COCO‡	CIDEr	145.3 (OFA)	**147.6** (+2.3)
	Finetuned Retrieval	COCO	R@1	72.5 (Florence)	**76.0** (+3.5)
		Flickr30K		92.6 (Florence)	**94.2** (+1.6)
	Zero-shot Retrieval	Flickr30K	R@1	86.5 (CoCa)	**88.2** (+1.7)

Table 1: Overview of BEIT-3 results on various vision and vision-language benchmarks. We compare with previous state-of-the-art models, including FD-SwinV2 [WHX+22], DINO [ZLL+22], Mask DINO [ZLL+22], FD-CLIP [WHX+22], CoCa [YWV+22], OFA [WYM+22], Florence [YCC+21]. We report the average of top-1 image-to-text and text-to-image results for retrieval tasks. "†" indicates ImageNet results only using publicly accessible resources. "‡" indicates image captioning results without CIDEr optimization.

1 Introduction: The Big Convergence

Recent years have featured a trend toward the big convergence of language [RNSS18, DCLT19, DYW+19], vision [BDPW22, PDB+22], and multimodal [WBDW21, RKH+21, YWV+22] pretraining. By performing large-scale pretraining on massive data, we can easily transfer the models to various downstream tasks. It is appealing that we can pretrain a general-purpose foundation model that handles multiple modalities. In this work, we advance the convergence trend for vision-language pretraining from the following three aspects.

First, the success of Transformers [VSP+17] is translated from language to vision [DBK+20] and multimodal [KSK21, WBDW21] problems. The unification of network architectures enables us to seamlessly handle multiple modalities. For vision-language modeling, there are various ways to apply Transformers due to the different natures of downstream tasks. For example, the dual-encoder architecture is used for efficient retrieval [RKH+21], encoder-decoder networks for generation tasks [WYY+21], and the fusion-encoder architecture for image-text encoding [KSK21]. However, most foundation models have to manually convert the end-task formats according to the specific architectures. Moreover, the parameters are usually not effectively shared across modalities. In this work, we adopt Multiway Transformers [WBDW21] for general-purpose modeling, i.e., one unified architecture shared for various downstream tasks. The modular network also comprehensively considers modality-specific encoding and cross-modality fusion.

Second, the pretraining task based on masked data modeling has been successfully applied to various modalities, such as texts [DCLT19], images [BDPW22, PDB+22], and image-text pairs [BWDW22]. Current vision-language foundation models usually multitask other pretraining objectives (such as image-text matching), rendering scaling-up unfriendly and inefficient. In contrast, we only use one pretraining task, i.e., mask-then-predict, to train a general-purpose multimodal foundation model. By regarding the image as a foreign language (i.e., *Imglish*), we handle texts and images in the same manner without fundamental modeling differences. Consequentially, image-text pairs are utilized as "parallel sentences" in order to learn the alignments between modalities. We also show that the simple yet effective method learns strong transferable representations, achieving state-of-the-art performance on both vision and vision-language tasks. The prominent success demonstrates the superiority of generative pretraining [DCLT19, BDPW22].

Third, scaling up the model size and data size universally improves the generalization quality of foundation models, so that we can transfer them to various downstream tasks. We follow the philosophy and scale up the model size to billions of parameters. Moreover, we scale up the pretraining data size in our experiments while only using publicly accessible resources for academic

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表 1：BEIT-3 在各种视觉和视觉-语言基准上的结果概览。我们与以往最先进的模型进行比较，包括 FD-SwinV2 [WHX+22]、DINO [ZLL+22]、Mask DINO [ZLL+22]、FD-CLIP [WHX+22]、CoCa [YWV+22]、OFA [WYM+22]、Florence [YCC+21]。对于检索任务，我们报告图像到文本和文本到图像 top-1 结果的平均值。“†”表示仅使用公开可访问资源的 ImageNet 结果。“‡”表示未进行 CIDEr 优化的图像描述生成结果。

1 引言：大融合

近年来，语言 [RNSS18, DCLT19, DYW+19]、视觉 [BDPW22, PDB+22] 和多模态 [WBDW21, RKH+21, YWV+22] 预训练呈现出大融合的趋势。通过在海量数据上进行大规模预训练，我们可以轻松地将模型迁移到各种下游任务。能够预训练一个处理多种模态的通用基础模型是很有吸引力的。在这项工作中，我们从以下三个方面推进视觉-语言预训练的融合趋势。

首先，Transformer [VSP+17] 的成功从语言领域迁移到了视觉 [DBK+20] 和多模态 [KSK21, WBDW21] 问题中。网络架构的统一使我们能够无缝处理多种模态。对于视觉-语言建模，由于下游任务的性质不同，应用 Transformer 的方式多种多样。例如，双编码器架构用于高效检索 [RKH+21]，编码器-解码器网络用于生成任务 [WYY+21]，而融合编码器架构用于图像-文本编码 [KSK21]。然而，大多数基础模型必须根据特定架构手动转换最终任务的格式。此外，参数通常无法在不同模态之间有效共享。在这项工作中，我们采用 Multiway Transformers [WBDW21] 进行通用建模，即用一种统一架构共享于各种下游任务。该模块化网络还全面考虑了模态特定编码和跨模态融合。

其次，基于掩码数据建模的预训练任务已成功应用于多种模态，例如文本 [DCLT19]、图像 [BDPW22, PDB+22] 以及图像-文本对 [BWDW22]。当前的视觉-语言基础模型通常会同时执行其他预训练目标（如图像-文本匹配），这使得规模扩展不友好且效率低下。相比之下，我们仅使用一个预训练任务，即先掩码后预测，来训练一个通用的多模态基础模型。通过将图像视为一种外语（即 *Imglish*），我们以相同的方式处理文本和图像，而不存在根本性的建模差异。因此，图像-文本对被用作“平行句子”，以学习模态之间的对齐。我们还表明，这种简单而有效的方法能够学习到强大的可迁移表示，并在视觉任务和视觉-语言任务上都取得了最先进的性能。这一显著成功证明了生成式预训练的优越性 [DCLT19, BDPW22]。

第三，扩大模型规模和数据规模通常能够提升基础模型的泛化质量，从而使我们能够将其迁移到各种下游任务中。我们遵循这一理念，将模型规模扩展到数十亿参数。此外，在实验中，我们扩大了预训练数据规模，同时仅使用公开可获取的学术资源。

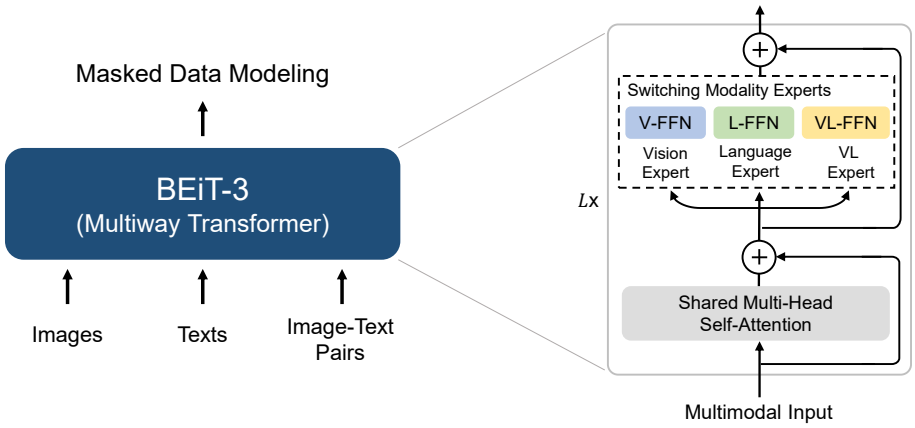


Figure 2: Overview of BEIT-3 pretraining. We perform masked data modeling on monomodal (i.e., images, and texts) and multimodal (i.e., image-text pairs) data with a shared Multiway Transformer as the backbone network.

reproducibility. Although without using any private data, our method outperforms state-of-the-art foundation models that rely on in-house data by a decent margin. In addition, the scaling up benefits from treating images as a foreign language, as we can directly reuse the pipeline developed for large-scale language model pretraining.

In this work, we take advantage of the above ideas to pretrain a general-purpose multimodal foundation model BEIT-3. We pretrain a Multiway Transformer by performing masked data modeling on images, texts, and image-text pairs. During pretraining, we randomly mask some proportion of text tokens or image patches. The self-supervised learning objective is to recover the original tokens (i.e., text tokens, or visual tokens) given corrupted inputs. The model is general-purpose in the sense that it can be repurposed for various tasks regardless of input modalities, or output formats.

As shown in Figure 1 and Table 1, BEIT-3 achieves state-of-the-art transfer performance across a broad range of vision and vision-language tasks. We evaluate BEIT-3 on extensive downstream tasks and datasets, i.e., object detection (COCO), instance segmentation (COCO), semantic segmentation (ADE20K), image classification (ImageNet), visual reasoning (NLVR2), visual question answering (VQAv2), image captioning (COCO), and cross-modal retrieval (Flickr30K, COCO). Specifically, our model outperforms previous strong foundation models [YWV+22, ADL+22, YCC+21] despite that we only use public resources for pretraining and finetuning. The model also obtains better results than specialized models. Moreover, BEIT-3 not only performs well on vision-language tasks but also on vision tasks (such as object detection, and semantic segmentation).

2 BEIT-3: A General-Purpose Multimodal Foundation Model

As shown in Figure 2, BEIT-3 is pretrained by masked data modeling on monomodal and multimodal data, using a shared Multiway Transformer network. The model can be transferred to various vision and vision-language downstream tasks.

2.1 Backbone Network: Multiway Transformers

We use Multiway Transformers [WBDW21] as the backbone model to encode different modalities. As shown in Figure 2, each Multiway Transformer block consists of a shared self-attention module, and a pool of feed-forward networks (i.e., modality experts) used for different modalities. We route each input token to the experts depending on its modality. In our implementation, each layer contains a vision expert and a language expert. Moreover, the top three layers have vision-language experts designed for fusion encoders. Refer to Figure 3 (a)(b)(c) for more detailed modeling layouts. Using a pool of modality experts encourages the model to capture more modality-specific information. The shared self-attention module learns the alignment between different modalities and enables deep fusion for multimodal (such as vision-language) tasks.

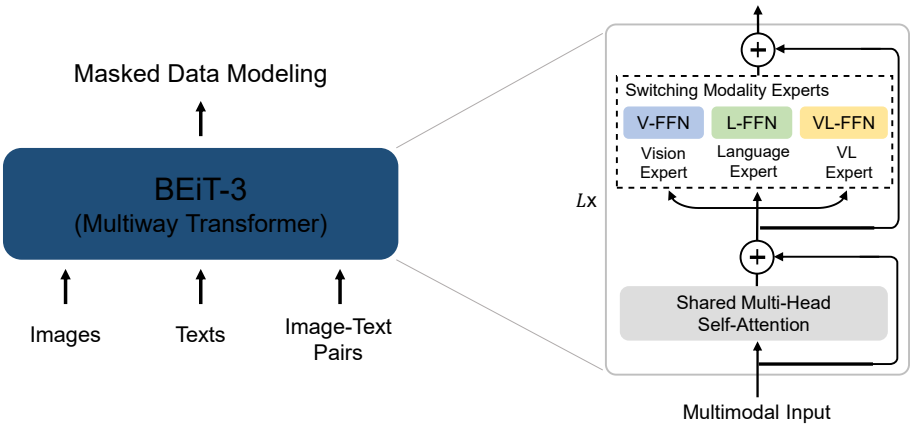


图2：BEIT-3 预训练概览。我们以共享的 Multiway Transformer 作为骨干网络，对单模态（即图像和文本）和多模态（即图像-文本对）数据进行掩码数据建模。

可复现性。尽管没有使用任何私有数据，我们的方法仍以相当可观的优势优于依赖内部数据的最先进基础模型。此外，将图像视为一门外语有利于规模扩展，因为我们可以直接复用为大规模语言模型预训练而开发的流程。

在这项工作中，我们利用上述思想来预训练一个通用多模态基础模型 BEIT-3。我们通过对图像、文本以及图文对进行掩码数据建模来预训练一个 Multiway Transformer。在预训练过程中，我们会随机掩盖一定比例的文本词元或图像块。自监督学习目标是在给定被破坏输入的情况下恢复原始词元（即文本词元或视觉词元）。该模型之所以是通用的，是因为无论输入模态或输出格式如何，它都可以被重新用于各种任务。

如图 1 和表 1 所示，BEIT-3 在广泛的视觉和视觉-语言任务中实现了最先进的迁移性能。我们在大量下游任务和数据集上评估了 BEIT-3，即目标检测（COCO）、实例分割（COCO）、语义分割（ADE20K）、图像分类（ImageNet）、视觉推理（NLVR2）、视觉问答（VQAv2）、图像描述生成（COCO）以及跨模态检索（Flickr30K、COCO）。具体而言，尽管我们仅使用公开资源进行预训练和微调，我们的模型仍优于以往强大的基础模型 [YWV+22, ADL+22, YCC+21]。该模型还取得了比专用模型更好的结果。此外，BEIT-3 不仅在视觉-语言任务上表现良好，也在视觉任务（如目标检测和语义分割）上表现出色。

2 BEIT-3：通用多模态基础模型

如图 2 所示，BEIT-3 使用共享的 Multiway Transformer 网络，通过在单模态和多模态数据上进行掩码数据建模来进行预训练。该模型可以迁移到各种视觉和视觉-语言下游任务。

2.1 主干网络：多路 Transformer

我们使用 Multiway Transformers [WBDW21] 作为骨干模型来编码不同模态。如图 2 所示，每个 Multiway Transformer 块由一个共享的自注意力模块和一组用于不同模态的前馈网络（即模态专家）组成。我们根据每个输入 token 的模态将其路由到相应的专家。在我们的实现中，每一层都包含一个视觉专家和一个语言专家。此外，最顶端的三层还包含专为融合编码器设计的视觉-语言专家。更详细的建模布局请参见图 3 (a)(b)(c)。使用一组模态专家有助于模型捕获更多模态特定的信息。共享的自注意力模块学习不同模态之间的对齐关系，并为多模态（如视觉-语言）任务实现深度融合。

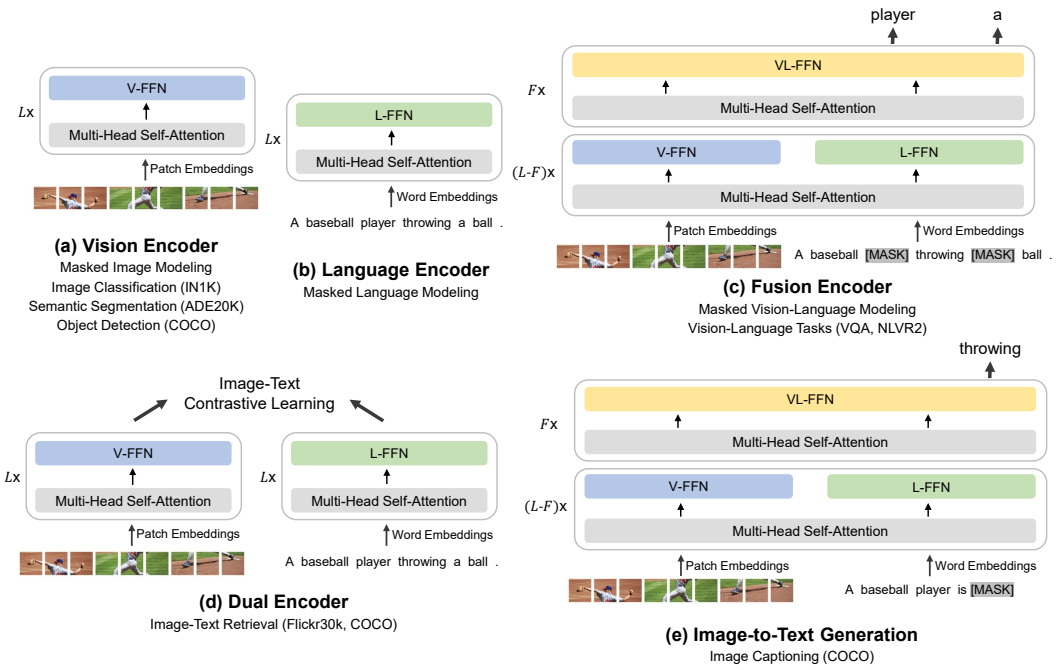


Figure 3: BEIT-3 can be transferred to various vision and vision-language downstream tasks. With a shared Multiway Transformer, we can reuse the model as (a)(b) vision or language encoders; (c) fusion encoders that jointly encode image-text pairs for deep interaction; (d) dual encoders that separately encode modalities for efficient retrieval; (e) sequence-to-sequence learning for image-to-text generation.

As shown in Figure 3, the unified architecture enables BEIT-3 to support a wide range of downstream tasks. For example, BEIT-3 can be used as an image backbone for various vision tasks, including image classification, object detection, instance segmentation, and semantic segmentation. It can also be finetuned as a dual encoder for efficient image-text retrieval, and a fusion model for multimodal understanding and generation tasks.

2.2 Pretraining Task: Masked Data Modeling

We pretrain BEIT-3 via a unified masked data modeling [BWDW22] objective on monomodal (i.e., images, and texts) and multimodal data (i.e., image-text pairs). During pretraining, we randomly mask some percentage of text tokens or image patches and train the model to recover the masked tokens. The unified mask-then-predict task not only learns representations but also learns the alignment of different modalities. Specifically, text data is tokenized by a SentencePiece tokenizer [KR18]. Image data is tokenized by the tokenizer of BEIT v2 [PDB+22] to obtain the discrete visual tokens as the reconstructed targets. We randomly mask 15% tokens of monomodal texts and 50% tokens of texts from image-text pairs. For images, we mask 40% of image patches using a block-wise masking strategy as in BEIT [BDPW22, PDB+22].

We only use one pretraining task, which makes the training process scaling-up friendly. In contrast, previous vision-language models [LYL+20, ZLH+21, KSK21, LSG+21, WBDW21, LLXH22, YWV+22] usually employ multiple pretraining tasks, such as image-text contrast, image-text matching, and word-patch/region alignment. We show that a much smaller pretraining batch size can be used with the mask-then-predict task. In comparison, contrastive-based models [RKH+21, JYX+21, YCC+21, YWV+22] usually need a very large batch size[2] for pretraining, which brings more engineering challenges, such as GPU memory cost.

[2]For example, CoCa [YWV+22] uses 65k batch size, CLIP [RKH+21] uses 32k batch size, and Florence [YCC+21] uses 24k batch size. BEIT-3 uses a much smaller 6k batch size for pretraining.

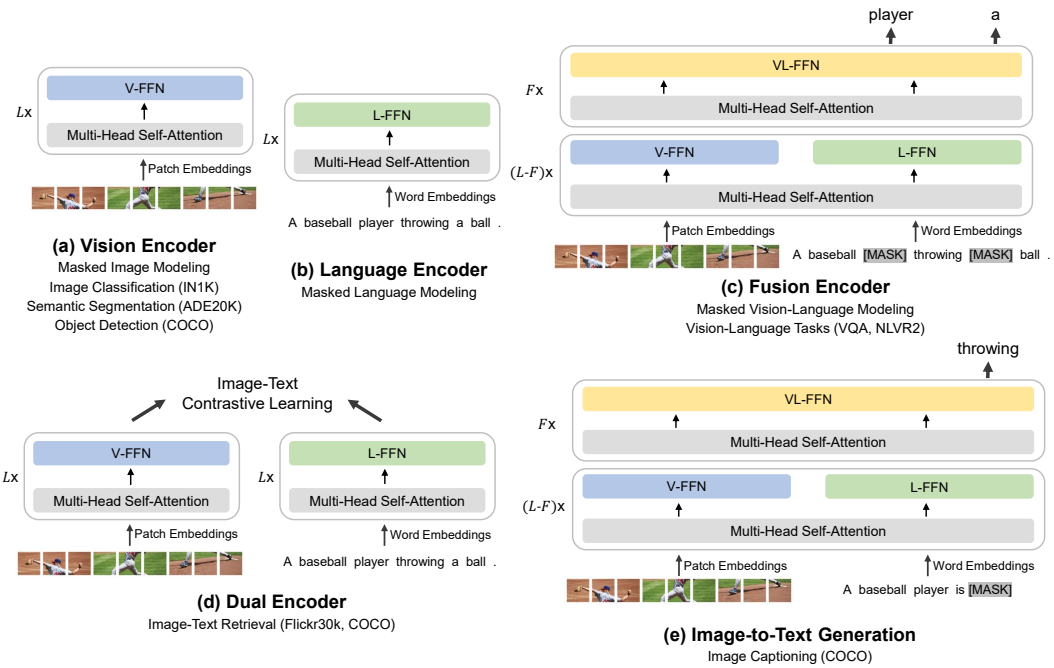


图 3：BEIT-3 可以迁移到各种视觉和视觉-语言下游任务。借助共享的 Multiway Transformer，我们可以将该模型复用为：(a)(b) 视觉或语言编码器；(c) 融合编码器，用于联合编码图像-文本对以进行深度交互；(d) 双编码器，用于分别编码不同模态以实现高效检索；(e) 序列到序列学习，用于图像到文本生成。

如图3所示，统一架构使 BEIT-3 能够支持广泛的下游任务。例如，BEIT-3 可作为各种视觉任务的图像骨干网络，包括图像分类、目标检测、实例分割和语义分割。它还可以被微调为双编码器，用于高效的图像-文本检索，以及作为融合模型用于多模态理解和生成任务。

2.2 预训练任务：掩码数据建模

我们通过在单模态数据（即图像和文本）以及多模态数据（即图文对）上采用统一的掩码数据建模 [BWDW22] 目标来预训练 BEIT-3。在预训练期间，我们随机掩蔽一定比例的文本标记或图像块，并训练模型恢复被掩蔽的标记。统一的“先掩蔽后预测”任务不仅学习表征，还学习不同模态之间的对齐。具体而言，文本数据由 SentencePiece 分词器 [KR18] 进行标记化。图像数据由 BEIT v2 [PDB+22] 的分词器进行标记化，以获得离散的视觉标记作为重建目标。我们随机掩蔽单模态文本中 15% 的标记，以及图文对中文本的 50% 标记。对于图像，我们采用与 BEIT [BDPW22, PDB+22] 中相同的按块掩蔽策略，掩蔽 40% 的图像块。

我们只使用一个预训练任务，这使得训练过程便于扩展。相比之下，先前的视觉-语言模型 [LYL+20, ZLH+21, KSK21, LSG+21, WBDW21, LLXH22, YWV+22] 通常采用多个预训练任务，例如图像-文本对比、图像-文本匹配以及词-图像块/区域对齐。我们表明，在遮蔽再预测任务中可以使用小得多的预训练批量大小。相比之下，基于对比学习的模型 [RKH+21, JYX+21, YCC+21, YWV+22] 通常需要非常大的批量大小[2]来进行预训练，这带来了更多工程挑战，例如 GPU 内存成本。

[2]For example, CoCa [YWV+22] uses 65k batch size, CLIP [RKH+21] uses 32k batch size, and Florence [YCC+21] uses 24k batch size. BEIT-3 uses a much smaller 6k batch size for pretraining.

Model	#Layers	Hidden Size	MLP Size	#Parameters				
				V-FFN	L-FFN	VL-FFN	Shared Attention	Total
BEIT-3	40	1408	6144	692M	692M	52M	317M	1.9B

Table 2: Model configuration of BEIT-3. The architecture layout follows ViT-giant [ZKHB21].

Data	Source	Size
Image-Text Pair	CC12M, CC3M, SBU, COCO, VG	21M pairs
Image	ImageNet-21K	14M images
Text	English Wikipedia, BookCorpus, OpenWebText, CC-News, Stories	160GB documents

Table 3: Pretraining data of BEIT-3. All the data are academically accessible.

2.3 Scaling Up: BEIT-3 Pretraining

Backbone Network BEIT-3 is a giant-size foundation model following the setup of ViT-giant [ZKHB21]. As shown in Table 2, the model consists of a 40-layer Multiway Transformer with 1408 hidden size, 6144 intermediate size, and 16 attention heads. All layers contain both vision experts and language experts. Vision-language experts are also employed in the top three Multiway Transformer layers. The self-attention module is shared across different modalities. BEIT-3 consists of 1.9B parameters in total, including 692M parameters for vision experts, 692M parameters for language experts, 52M parameters for vision-language experts, and 317M parameters for the shared self-attention module. Notice that only vision-related parameters (i.e., comparable size as ViT-giant; about 1B) are activated when the model is used as a vision encoder.

Pretraining Data BEIT-3 is pretrained on both monomodal and multimodal data shown in Table 3. For multimodal data, there are about 15M images and 21M image-text pairs collected from five public datasets: Conceptual 12M (CC12M) [CSDS21], Conceptual Captions (CC3M) [SDGS18], SBU Captions (SBU) [OKB11], COCO [LMB+14] and Visual Genome (VG) [KZG+17]. For monomodal data, we use 14M images from ImageNet-21K and 160GB text corpora [BDW+20] from English Wikipedia, BookCorpus [ZKZ+15], OpenWebText[3], CC-News [LOG+19], and Stories [TL18].

Pretraining Settings We pretrain BEIT-3 for 1M steps. Each batch contains 6144 samples in total, including 2048 images, 2048 texts and 2048 image-text pairs. The batch size is much smaller than contrastive models [RKH+21, JYX+21, YWV+22]. BEIT-3 uses 14×14 patch size and is pretrained at resolution 224×224. We use the same image augmentation as in BEIT [BDPW22], including random resized cropping, horizontal flipping, and color jittering [WXYL18]. A SentencePiece tokenizer [KR18] with 64k vocab size is employed to tokenize the text data. We use the AdamW [LH19] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\epsilon =$1e-6 for optimization. We use a cosine learning rate decay scheduler with a peak learning rate of 1e-3 and a linear warmup of 10k steps. The weight decay is 0.05. Stochastic depth [HSL+16] with a rate of 0.1 is used. The BEIT initialization algorithm[4] [BDPW22] is used to stabilize Transformer training.

3 Experiments on Vision and Vision-Language Tasks

We extensively evaluate BEIT-3 on major public benchmarks for both vision-language and vision tasks. Table 1 presents the overview of results. BEIT-3 obtains state-of-the-art performance on a wide range of vision and vision-language tasks.

[3] http://skylion007.github.io/OpenWebTextCorpus

[4] We first randomly initialize the parameters within a small range, e.g., $[-0.02, 0.02]$. Next, we rescale the l-th Transformer layer's output matrices (i.e., the last linear projection within each sublayer) of self-attention and FFN by $\frac{1}{\sqrt{2l}}$.

Model	#Layers	Hidden Size	MLP Size	#Parameters				
				V-FFN	L-FFN	VL-FFN	Shared Attention	Total
BEIT-3	40	1408	6144	692M	692M	52M	317M	1.9B

表 2：BEIT-3 的模型配置。架构布局遵循 ViT-giant [ZKHB21]。

Data	Source	Size
Image-Text Pair	CC12M, CC3M, SBU, COCO, VG	21M pairs
Image	ImageNet-21K	14M images
Text	English Wikipedia, BookCorpus, OpenWebText, CC-News, Stories	160GB documents

表 3：BEIT-3 的预训练数据。所有数据均可通过学术途径获取。

2.3 扩展规模：BEIT-3 预训练

骨干网络 BEIT-3 是一个遵循 ViT-giant [ZKHB21] 设置的巨型基础模型。如表 2 所示，该模型由一个 40 层的 Multiway Transformer 组成，隐藏层维度为 1408，中间层维度为 6144，并具有 16 个注意力头。所有层都包含视觉专家和语言专家。在最上面的三层 Multiway Transformer 中还采用了视觉-语言专家。自注意力模块在不同模态之间共享。BEIT-3 总共包含 1.9B 个参数，其中视觉专家有 692M 个参数，语言专家有 692M 个参数，视觉-语言专家有 52M 个参数，共享自注意力模块有 317M 个参数。请注意，当该模型用作视觉编码器时，仅激活与视觉相关的参数（即规模与 ViT-giant 相当；约 1B）。

预训练数据 BEIT-3 在表 3 所示的单模态和多模态数据上进行预训练。对于多模态数据，包含约 1500 万张图像和 2100 万个图像-文本对，收集自五个公开数据集：Conceptual 12M (CC 12M) [CSDS21]、Conceptual Captions (CC3M) [SDGS18]、SBU Captions (SBU) [OKB11]、COCO [LMB+14] 和 Visual Genome (VG) [KZG+17]。对于单模态数据，我们使用来自 ImageNet-21K 的 1400 万张图像，以及来自英文维基百科、BookCorpus [ZKZ+15]、OpenWebText[3]、CC-News [LOG+19] 和 Stories [TL18] 的 160GB 文本语料库 [BDW+20]。

预训练设置 我们对 BEIT-3 进行 1M 步的预训练。每个批次总共包含 6144 个样本，包括 2048 张图像、2048 段文本和 2048 个图像-文本对。其批量大小远小于对比式模型 [RKH+21, JYX+21, YWV+22]。BEIT-3 使用 14×14 的 patch 大小，并在 224×224 的分辨率下进行预训练。我们使用与 BEIT [BDPW22] 相同的图像增强方法，包括随机缩放裁剪、水平翻转和颜色抖动 [WXYL18]。采用词表大小为 64k 的 SentencePiece 分词器 [KR18] 对文本数据进行分词。我们使用 AdamW [LH19] 优化器进行优化，其中 $\beta_1 = 0.9$、$\beta_2 = 0.98$，且 ϵ =1e-6。我们使用余弦学习率衰减调度器，峰值学习率为 1e-3，并进行 10k 步的线性预热。权重衰减为 0.05。使用随机深度 [HSL+16]，其比率为 0.1。采用 BEiT 初始化算法[4] [BDPW22] 来稳定 Transformer 训练。

3 视觉和视觉-语言任务实验

我们在主要的公开基准上对 BEIT-3 进行了广泛评估，涵盖视觉-语言任务和视觉任务。表 1 给出了结果概览。BEIT-3 在广泛的视觉和视觉-语言任务上取得了最先进的性能。

[3] `http://skylion007.github.io/OpenWebTextCorpus`

[4] We first randomly initialize the parameters within a small range, e.g., $[-0.02, 0.02]$. Next, we rescale the l-th Transformer layer's output matrices (i.e., the last linear projection within each sublayer) of self-attention and FFN by $\frac{1}{\sqrt{2l}}$.

Model	VQAv2		NLVR2		COCO Captioning			
	test-dev	test-std	dev	test-P	B@4	M	C	S
Oscar [LYL+20]	73.61	73.82	79.12	80.37	37.4	30.7	127.8	23.5
VinVL [ZLH+21]	76.52	76.60	82.67	83.98	38.5	30.4	130.8	23.4
ALBEF [LSG+21]	75.84	76.04	82.55	83.14	-	-	-	-
BLIP [LLXH22]	78.25	78.32	82.15	82.24	40.4	-	136.7	-
SimVLM [WYY+21]	80.03	80.34	84.53	85.15	40.6	33.7	143.3	**25.4**
Florence [YCC+21]	80.16	80.36	-	-	-	-	-	-
OFA [WYM+22]	82.00	82.00	-	-	43.9	31.8	145.3	24.8
Flamingo [ADL+22]	82.00	82.10	-	-	-	-	138.1	-
CoCa [YWV+22]	82.30	82.30	86.10	87.00	40.9	**33.9**	143.6	24.7
BEIT-3	**84.19**	**84.03**	**91.51**	**92.58**	**44.1**	32.4	**147.6**	**25.4**

Table 4: Results of visual question answering, visual reasoning, and image captioning tasks. We report *vqa-score* on VQAv2 test-dev and test-standard splits, accuracy for NLVR2 development set and public test set (test-P). For COCO image captioning, we report BLEU@4 (B@4), METEOR (M), CIDEr (C), and SPICE (S) on the Karpathy test split. For simplicity, we report captioning results without using CIDEr optimization.

3.1 Vision-Language Downstream Tasks

We evaluate the capabilities of BEIT-3 on the widely used vision-language understanding and generation benchmarks, including visual question answering [GKS+17], visual reasoning [SZZ+19], image-text retrieval [PWC+15, LMB+14], and image captioning [LMB+14].

Visual Question Answering (VQA) The task requires the model to answer natural language questions about input images. Following previous work [AHB+18, ZLH+21, KSK21], we conduct finetuning experiments on the VQA v2.0 dataset [GKS+17] and formulate the task as a classification problem. The model is trained to predict answers from the 3129 most frequent answer candidates in the training set. BEIT-3 is finetuned as a fusion encoder to model deep interactions of images and questions for the VQA task. We concatenate the embeddings of a given question and an image, and then feed the input embeddings into Multiway Transformers to jointly encode the image-question pair. The final pooled output is fed into a classifier layer to predict the answer. The results are present in Table 4, BEIT-3 outperforms all previous models by a large margin (more than 1.7 points), pushing the state of the art to 84.03 with a single model.

Visual Reasoning The task needs models to perform joint reasoning about images and natural language descriptions. We evaluate the model on the popular NLVR2 [SZZ+19] benchmark, which is to determine whether a textual description is true about a pair of images. Following previous work [ZLH+21, KSK21], we construct two image-text pairs based on the triplet input. We finetune BEIT-3 as a fusion encoder to jointly encode the image-text pairs. The final pooled outputs of the two pairs are concatenated and then fed into a classifier layer to predict the label. As shown in Table 4, BEIT-3 achieves a new state-of-the-art result for visual reasoning, outperforming CoCa by about 5.6 points. The performance on NLVR2 reaches above 90% for the first time.

Image Captioning The task aims to generate a natural language caption for the given image. We use the COCO [LMB+14] benchmark, finetune and evaluate the model on Karpathy split [KF15]. Following UNILM [DYW+19] and s2s-ft [BDW+21], BEIT-3 is used as a conditional generation model via masked finetuning. To be more specific, a special self-attention mask is employed for the image captioning task. Image tokens (i.e., image patches) can only attend to each other bidirectionally within the image sequence. Tokens of the caption can attention to image tokens, their leftward caption tokens, and themselves. During finetuning, we randomly mask some percentage of caption tokens. The model is trained to recover these tokens based on the clues of the image and its leftward caption context. We also mask the special boundary token `[SEP]` to help the model learn to terminate the generation. For simplicity, BEIT-3 is trained with simple cross-entropy loss, without using CIDEr optimization. During inference, we generate the caption tokens one by one in an autoregressive

Model	VQAv2		NLVR2		COCO Captioning			
	test-dev	test-std	dev	test-P	B@4	M	C	S
Oscar [LYL+20]	73.61	73.82	79.12	80.37	37.4	30.7	127.8	23.5
VinVL [ZLH+21]	76.52	76.60	82.67	83.98	38.5	30.4	130.8	23.4
ALBEF [LSG+21]	75.84	76.04	82.55	83.14	-	-	-	-
BLIP [LLXH22]	78.25	78.32	82.15	82.24	40.4	-	136.7	-
SimVLM [WYY+21]	80.03	80.34	84.53	85.15	40.6	33.7	143.3	**25.4**
Florence [YCC+21]	80.16	80.36	-	-	-	-	-	-
OFA [WYM+22]	82.00	82.00	-	-	43.9	31.8	145.3	24.8
Flamingo [ADL+22]	82.00	82.10	-	-	-	-	138.1	-
CoCa [YWV+22]	82.30	82.30	86.10	87.00	40.9	**33.9**	143.6	24.7
BEIT-3	**84.19**	**84.03**	**91.51**	**92.58**	**44.1**	32.4	**147.6**	**25.4**

表 4：视觉问答、视觉推理和图像描述任务的结果。我们报告了 VQAv2 test-dev 和 test-standard 划分上的 *vqa-score*，以及 NLVR2 开发集和公开测试集（test-P）上的准确率。对于 COCO 图像描述，我们报告 Karpathy 测试划分上的 BLEU@4（B@4）、METEOR（M）、CIDEr（C）和 SPICE（S）。为简化起见，我们报告未使用 CIDEr 优化的图像描述结果。

3.1 视觉-语言下游任务

我们在广泛使用的视觉-语言理解与生成基准上评估 BEIT-3 的能力，包括视觉问答 [GKS+17]、视觉推理 [SZZ+19]、图文检索 [PWC+15, LMB+14] 和图像描述生成 [LMB+14]。

视觉问答（VQA）该任务要求模型回答关于输入图像的自然语言问题。遵循以往工作 [AHB+18, ZLH+21, KSK21]，我们在 VQA v2.0 数据集 [GKS+17] 上进行微调实验，并将该任务表述为一个分类问题。模型经过训练，从训练集中最常见的 3129 个候选答案中预测答案。BEIT-3 被微调为一个融合编码器，用于为 VQA 任务建模图像与问题之间的深层交互。我们将给定问题和图像的嵌入进行拼接，然后将输入嵌入送入 Multiway Transformers，以联合编码图像-问题对。最终的池化输出被输入到分类器层以预测答案。结果见表 4，BEIT-3 以较大优势（超过 1.7 个点）优于所有先前模型，将单模型的最新水平提升至 84.03。

视觉推理 该任务需要模型对图像和自然语言描述进行联合推理。我们在流行的 NLVR2 [SZZ+19] 基准上评估模型，该基准旨在判断一段文本描述对于一对图像是否为真。遵循先前工作 [ZLH+21, KSK21]，我们基于三元组输入构建两个图文对。我们将 BEIT-3 微调为融合编码器，以联合编码图文对。两个图文对的最终池化输出被拼接在一起，然后输入分类器层以预测标签。如表 4 所示，BEIT-3 在视觉推理任务上取得了新的最先进结果，比 CoCa 高出约 5.6 个点。NLVR2 上的性能首次超过 90%。

图像描述生成 该任务旨在为给定图像生成自然语言描述。我们使用 COCO [LMB+14] 基准，并在 Karpathy 划分 [KF15] 上对模型进行微调和评估。遵循 UNILM [DYW+19] 和 s2s-ft [BDW+21]，BEIT-3 通过掩码微调被用作条件生成模型。更具体地说，图像描述生成任务采用了一种特殊的自注意力掩码。图像 token（即图像块）只能在图像序列内部彼此进行双向注意力。描述文本的 token 可以关注图像 token、其左侧的描述 token 以及自身。在微调过程中，我们随机掩码一定比例的描述 token。模型被训练为基于图像线索及其左侧描述上下文来恢复这些 token。我们还会掩码特殊边界 token [SEP]，以帮助模型学习终止生成。为简单起见，BEIT-3 使用简单的交叉熵损失进行训练，而不使用 CIDEr 优化。在推理过程中，我们以自回归方式逐个生成描述 token

Model	MSCOCO (5K test set)						Flickr30K (1K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Fusion-encoder models												
UNITER [CLY+20]	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA [GCL+20]	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
Oscar [LYL+20]	73.5	92.2	96.0	57.5	82.8	89.8	-	-	-	-	-	-
VinVL [ZLH+21]	75.4	92.9	96.2	58.8	83.5	90.3	-	-	-	-	-	-
Dual encoder + Fusion encoder reranking												
ALBEF [LSG+21]	77.6	94.3	97.2	60.7	84.3	90.5	95.9	99.8	**100.0**	85.6	97.5	98.9
BLIP [LLXH22]	82.4	95.4	97.9	65.1	86.3	91.8	97.4	99.8	99.9	87.6	97.7	99.0
Dual-encoder models												
ALIGN [JYX+21]	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	**100.0**	84.9	97.4	98.6
FILIP [YHH+21]	78.9	94.4	97.4	61.2	84.3	90.6	96.6	**100.0**	**100.0**	87.1	97.7	99.1
Florence [YCC+21]	81.8	95.2	-	63.2	85.7	-	97.2	99.9	-	87.9	98.1	-
BEIT-3	**84.8**	**96.5**	**98.3**	**67.2**	**87.7**	**92.8**	**98.0**	**100.0**	**100.0**	**90.3**	**98.7**	**99.5**

Table 5: Finetuning results of image-to-text retrieval and text-to-image retrieval on COCO and Flickr30K. Notice that dual-encoder models are more efficient than fusion-encoder-based models for the retrieval tasks.

Model	Flickr30K (1K test set)					
	Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10
FLAVA [SHG+21]	67.7	94.0	-	65.2	89.4	-
CLIP [RKH+21]	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN [JYX+21]	88.6	98.7	99.7	75.7	93.8	96.8
FILIP [YHH+21]	89.8	99.2	99.8	75.0	93.4	96.3
Florence [YCC+21]	90.9	99.1	-	76.7	93.6	-
Flamingo [ADL+22]	89.3	98.8	99.7	79.5	95.3	**97.9**
CoCa [YWV+22]	92.5	99.5	99.9	80.4	**95.7**	97.7
BEIT-3	**94.9**	**99.9**	**100.0**	**81.5**	95.6	97.8

Table 6: Zero-shot image-to-text retrieval and text-to-image retrieval on Flickr30K.

manner. Table 4 presents the results on COCO captioning. BEIT-3 outperforms all previous models trained with cross-entropy loss, creating a new state-of-the-art image captioning result. The results demonstrate the superiority of BEIT-3 for vision-language generation.

Image-Text Retrieval The task is to measure the similarity between images and texts. There are two directions depending on the modality of the retrieved target: image-to-text retrieval, and text-to-image retrieval. Two popular retrieval benchmarks, i.e., COCO [LMB+14], and Flickr30K [PWC+15], are used to evaluate the model. Following previous work [ZLH+21, KSK21], we use the Karpathy split [KF15] for the two benchmarks. BEIT-3 is finetuned as a dual encoder for efficient image-text retrieval. Dual-encoder models separately encode images and texts to obtain their representations. Then we calculate the cosine similarity scores of these representations. Dual-encoder models are more efficient than fusion-encoder models. Because they do not have to jointly encode all possible image-text pairs.

We directly finetune BEIT-3 on COCO and Flickr30K, although the model is not pretrained with image-text contrastive loss. Surprisingly, BEIT-3 outperforms previous state-of-the-art models only using a small amount of contrastive training. The results demonstrate that BEIT-3 effectively learns alignments between images and texts via masked data modeling. In order to improve the performance, we perform intermediate finetuning with an image-text contrastive objective on the pretraining image-text pairs. We finetune the model with much fewer steps than pretraining. Then we use the model to evaluate zero-shot and finetuned image-text retrieval. The finetuned results are present

Model	MSCOCO (5K test set)						Flickr30K (1K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Fusion-encoder models												
UNITER [CLY+20]	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA [GCL+20]	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
Oscar [LYL+20]	73.5	92.2	96.0	57.5	82.8	89.8	-	-	-	-	-	-
VinVL [ZLH+21]	75.4	92.9	96.2	58.8	83.5	90.3	-	-	-	-	-	-
Dual encoder + Fusion encoder reranking												
ALBEF [LSG+21]	77.6	94.3	97.2	60.7	84.3	90.5	95.9	99.8	**100.0**	85.6	97.5	98.9
BLIP [LLXH22]	82.4	95.4	97.9	65.1	86.3	91.8	97.4	99.8	99.9	87.6	97.7	99.0
Dual-encoder models												
ALIGN [JYX+21]	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	**100.0**	84.9	97.4	98.6
FILIP [YHH+21]	78.9	94.4	97.4	61.2	84.3	90.6	96.6	**100.0**	**100.0**	87.1	97.7	99.1
Florence [YCC+21]	81.8	95.2	-	63.2	85.7	-	97.2	99.9	-	87.9	98.1	-
BEIT-3	**84.8**	**96.5**	**98.3**	**67.2**	**87.7**	**92.8**	**98.0**	**100.0**	**100.0**	**90.3**	**98.7**	**99.5**

表 5：在 COCO 和 Flickr30K 上进行图像到文本检索和文本到图像检索的微调结果。请注意，对于检索任务，双编码器模型比基于融合编码器的模型更高效。

Model	Flickr30K (1K test set)					
	Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10
FLAVA [SHG+21]	67.7	94.0	-	65.2	89.4	-
CLIP [RKH+21]	88.0	98.7	99.4	68.7	90.6	95.2
ALIGN [JYX+21]	88.6	98.7	99.7	75.7	93.8	96.8
FILIP [YHH+21]	89.8	99.2	99.8	75.0	93.4	96.3
Florence [YCC+21]	90.9	99.1	-	76.7	93.6	-
Flamingo [ADL+22]	89.3	98.8	99.7	79.5	95.3	**97.9**
CoCa [YWV+22]	92.5	99.5	99.9	80.4	**95.7**	97.7
BEIT-3	**94.9**	**99.9**	**100.0**	**81.5**	95.6	97.8

表 6：在 Flickr30K 上的零样本图像到文本检索和文本到图像检索。

方式。表4展示了在 COCO 图像描述任务上的结果。BEIT-3 优于所有先前使用交叉熵损失训练的模型，创造了新的最先进图像描述结果。这些结果证明了 BEIT-3 在视觉-语言生成方面的优越性。

图文检索 该任务旨在衡量图像与文本之间的相似性。根据检索目标的模态不同，有两个方向：图像到文本检索和文本到图像检索。我们使用两个常用的检索基准，即 COCO [LMB+14] 和 Flickr30K [PWC+15]，来评估模型。遵循以往工作 [ZLH+21, KSK21]，我们在这两个基准上使用 Karpathy 划分 [KF15]。BEIT-3 被微调为双编码器，以实现高效的图文检索。双编码器模型分别对图像和文本进行编码，以获得它们的表示。然后我们计算这些表示的余弦相似度分数。双编码器模型比融合编码器模型更高效，因为它们不必联合编码所有可能的图文对。

我们直接在 COCO 和 Flickr30K 上对 BEIT-3 进行微调，尽管该模型并未使用图文对比损失进行预训练。令人惊讶的是，BEIT-3 仅使用少量对比训练就超越了以往的最先进模型。结果表明，BEIT-3 通过掩码数据建模有效地学习了图像与文本之间的对齐关系。为了提升性能，我们在预训练的图文对上使用图文对比目标进行中间微调。我们使用比预训练少得多的步数对模型进行微调。然后，我们使用该模型评估零样本和微调后的图文检索。微调结果如下。

Model	Extra OD Data	Maximum Image Size	COCO test-dev AP^{box}	COCO test-dev AP^{mask}
ViT-Adapter [CDW+22]	-	1600	60.1	52.1
DyHead [DCX+21]	ImageNet-Pseudo Labels	2000	60.6	-
Soft Teacher [XZH+21]	Object365	-	61.3	53.0
GLIP [LZZ+21]	FourODs	-	61.5	-
GLIPv2 [ZZH+22]	FourODs	-	62.4	-
Florence [YCC+21]	FLOD-9M	2500	62.4	-
SwinV2-G [LHL+21]	Object365	1536	63.1	54.4
Mask DINO [LZX+22]	Object365	1280	-	54.7
DINO [ZLL+22]	Object365	2000	63.3	-
BEIT-3	Object365	1280	**63.7**	**54.8**

Table 7: Results of object detection and instance segmentation on COCO benchmark. BEIT-3 uses Cascade Mask R-CNN [CV21] as the detection head. Our results are reported with multi-scale evaluation. We report the maximum image size used for training. FLOD-9M and FourODs also contain Object365. The results of the comparison systems are from the paperswithcode.com leaderboard (timestamp: 08/22/2022).

in Table 5, dual-encoder BEIT-3 outperforms prior models by a large margin, achieving 3.0/4.0 absolute improvement on COCO top-1 image-to-text/text-to-image retrieval, and 0.8/2.4 absolute improvement on Flickr30K top-1 image-to-text/text-to-image retrieval. BEIT-3 also significantly outperforms fusion-encoder-based models, which require more computation cost for inference. As present in Table 6, BEIT-3 also achieves better performance than previous models on Flickr30K zero-shot retrieval.

3.2 Vision Downstream Tasks

In addition to vision-language downstream tasks, BEIT-3 can be transferred to a wide range of vision downstream tasks, including object detection, instance segmentation, semantic segmentation, and image classification. The number of effective parameters is comparable to ViT-giant [ZKHB21], i.e., about 1B, when BEIT-3 is used as a vision encoder.

Object Detection and Instance Segmentation We conduct finetuning experiments on the COCO 2017 benchmark [LMB+14], which consists of 118k training, 5k validation, and 20k test-dev images. We use BEIT-3 as the backbone and follow ViTDet [LMGH22], including a simple feature pyramid and window attention, for the object detection and instance segmentation tasks. Following common practices [LHL+21, ZLL+22], we first conduct intermediate finetuning on the Objects365 [SLZ+19] dataset. Then we finetune the model on the COCO dataset. Soft-NMS [BSCD17] is used during inference. Table 7 compares BEIT-3 with previous state-of-the-art models on COCO object detection and instance segmentation. BEIT-3 achieves the best results on the COCO test-dev set with a smaller image size used for finetuning, reaching up to 63.7 box AP and 54.8 mask AP.

Semantic Segmentation Semantic segmentation aims to predict the label for each pixel of the given image. We evaluate BEIT-3 on the challenging ADE20K dataset [ZZP+19], which includes 150 semantic categories. ADE20K contains 20k images for training and 2k images for validation. We directly follow the task transfer settings of ViT-Adapter [CDW+22]. We use a dense prediction task adapter and employ Mask2Former [CMS+21] as the segmentation framework. As shown in Table 8, BEIT-3 creates a new state-of-the-art result with 62.8 mIoU, outperforming FD-SwinV2 [WHX+22] giant model with 3B parameters by 1.4 points. It shows that BEIT-3 achieves superior performance on the dense prediction task.

Image Classification We evaluate the model on ImageNet-1K [RDS+15], which contains 1.28M training images and 50k validation images in 1k classes. Rather than appending a task layer to the vision encoder [DBK+20, BDPW22], we formulate the task as an image-to-text retrieval task. We use the category names as texts to construct image-text pairs. BEIT-3 is trained as a dual encoder to find the most relevant label for an image. During inference, we first compute the feature embeddings

Model	Extra OD Data	Maximum Image Size	COCO test-dev AP^{box}	COCO test-dev AP^{mask}
ViT-Adapter [CDW+22]	-	1600	60.1	52.1
DyHead [DCX+21]	ImageNet-Pseudo Labels	2000	60.6	-
Soft Teacher [XZH+21]	Object365	-	61.3	53.0
GLIP [LZZ+21]	FourODs	-	61.5	-
GLIPv2 [ZZH+22]	FourODs	-	62.4	-
Florence [YCC+21]	FLOD-9M	2500	62.4	-
SwinV2-G [LHL+21]	Object365	1536	63.1	54.4
Mask DINO [LZX+22]	Object365	1280	-	54.7
DINO [ZLL+22]	Object365	2000	63.3	-
BEIT-3	Object365	1280	**63.7**	**54.8**

表 7：COCO 基准上目标检测和实例分割的结果。BEIT-3 使用 Cascade Mask R-CNN [CV21] 作为检测头。我们的结果采用多尺度评估报告。我们报告了训练中使用的最大图像尺寸。FLOD-9M 和 FourODs 也包含 Object365。对比系统的结果来自 paperswithcode.com 排行榜（时间戳：2022 年 08 月 22 日）。

在表 5 中，双编码器 BEIT-3 大幅优于以往模型，在 COCO top-1 图像到文本/文本到图像检索上实现了 3.0/4.0 的绝对提升，在 Flickr30K top-1 图像到文本/文本到图像检索上实现了 0.8/2.4 的绝对提升。BEIT-3 也显著优于基于融合编码器的模型，后者在推理时需要更高的计算成本。如表 6 所示，BEIT-3 在 Flickr30K 零样本检索上也取得了比以往模型更好的性能。

3.2 视觉下游任务

除了视觉-语言下游任务之外，BEIT-3 还可以迁移到广泛的视觉下游任务，包括目标检测、实例分割、语义分割和图像分类。当 BEIT-3 用作视觉编码器时，其有效参数数量与 ViT-giant [ZKHB21] 相当，即约 10 亿。

目标检测与实例分割 我们在 COCO 2017 基准 [LMB+14] 上进行了微调实验，该基准包含 118k 张训练图像、5k 张验证图像和 20k 张 test-dev 图像。我们使用 BEIT-3 作为骨干网络，并遵循 ViTDet [LMGH22]，包括简单的特征金字塔和窗口注意力，用于目标检测和实例分割任务。按照常见做法 [LHL+21, ZLL+22]，我们首先在 Objects365 [SLZ+19] 数据集上进行中间微调。然后我们在 COCO 数据集上对模型进行微调。推理过程中使用 Soft-NMS [BSCD17]。表 7 将 BEIT-3 与此前在 COCO 目标检测和实例分割上的最先进模型进行了比较。BEIT-3 在用于微调的图像尺寸更小的情况下，在 COCO test-dev 集上取得了最佳结果，达到 63.7 的 box AP 和 54.8 的 mask AP。

语义分割 语义分割旨在预测给定图像中每个像素的标签。我们在具有挑战性的 ADE20K 数据集 [ZZP+19] 上评估 BEIT-3，该数据集包含 150 个语义类别。ADE20K 包含 2 万张训练图像和 2 千张验证图像。我们直接遵循 ViT-Adapter [CDW+22] 的任务迁移设置。我们使用密集预测任务适配器，并采用 Mask2Former [CMS+21] 作为分割框架。如表 8 所示，BEIT-3 以 62.8 mIoU 创造了新的最先进结果，比拥有 30 亿参数的 FD-SwinV2 [WHX+22] 巨型模型高出 1.4 个点。这表明 BEIT-3 在密集预测任务上实现了卓越性能。

图像分类 我们在 ImageNet-1K [RDS+15] 上评估模型，该数据集包含 1k 个类别中的 1.28M 张训练图像和 50k 张验证图像。我们没有像 [DBK+20, BDPW22] 那样在视觉编码器后附加一个任务层，而是将该任务表述为图像到文本检索任务。我们使用类别名称作为文本来构建图像-文本对。BEIT-3 被训练为双编码器，以找到与图像最相关的标签。在推理过程中，我们首先计算特征嵌入

Model	Crop Size	ADE20K	
		mIoU	+MS
HorNet [RZT+22]	640^2	57.5	57.9
SeMask [JSO+21]	640^2	57.0	58.3
SwinV2-G [LHL+21]	896^2	59.3	59.9
ViT-Adapter [CDW+22]	896^2	59.4	60.5
Mask DINO [LZX+22]	-	59.5	60.8
FD-SwinV2-G [WHX+22]	896^2	-	61.4
BEIT-3	896^2	**62.0**	**62.8**

Table 8: Results of semantic segmentation on ADE20K. "MS" is short for multi-scale. The results of the comparison systems are from the paperswithcode.com leaderboard (timestamp: 08/22/2022).

Model	Extra Data	Image Size	ImageNet
*With extra **private** image-tag data*			
SwinV2-G [LHL+21]	IN-22K-ext-70M	640^2	90.2
ViT-G [ZKHB21]	JFT-3B	518^2	90.5
CoAtNet-7 [DLLT21]	JFT-3B	512^2	90.9
Model Soups [WIG+22]	JFT-3B	500^2	91.0
CoCa [YWV+22]	JFT-3B	576^2	91.0
*With only **public** image-tag data*			
BEIT [BDPW22]	IN-21K	512^2	88.6
CoAtNet-4 [DLLT21]	IN-21K	512^2	88.6
MaxViT [TTZ+22]	IN-21K	512^2	88.7
MViTv2 [LWF+22]	IN-21K	512^2	88.8
FD-CLIP [WHX+22]	IN-21K	336^2	89.0
BEIT-3	IN-21K	336^2	**89.6**

Table 9: Top-1 accuracy on ImageNet-1K.

of possible class names and the feature embedding of the image. Their cosine similarity scores are then calculated to predict the most probable label for each image. Table 9 reports the results on ImageNet-1K. We first perform intermediate finetuning on ImageNet-21K, then we train the model on ImageNet-1K. For a fair comparison, we compare with the previous models only using public image-tag data. BEIT-3 outperforms prior models, creating a new state-of-the-art result when only using public image-tag data.

4 Conclusion

In this paper, we present BEIT-3, a general-purpose multimodal foundation model, which achieves state-of-the-art performance across a wide range of vision and vision-language benchmarks. The key idea of BEIT-3 is that image can be modeled as a foreign language, so that we can conduct masked "language" modeling over images, texts, and image-text pairs in a unified way. We also demonstrate that Multiway Transformers can effectively model different vision and vision-language tasks, making it an intriguing option for general-purpose modeling. BEIT-3 is simple and effective, and is a promising direction for scaling up multimodal foundation models. For future work, we are working on pretraining multilingual BEIT-3 and including more modalities (e.g., audio) in BEIT-3 to facilitate the cross-lingual and cross-modality transfer, and advance the big convergence of large-scale pretraining across tasks, languages, and modalities. We are also interested in enabling in-context learning capability for multimodal foundation models by combining the strength of BEIT-3 and MetaLM [HSD+22].

Model	Crop Size	ADE20K	
		mIoU	+MS
HorNet [RZT+22]	640^2	57.5	57.9
SeMask [JSO+21]	640^2	57.0	58.3
SwinV2-G [LHL+21]	896^2	59.3	59.9
ViT-Adapter [CDW+22]	896^2	59.4	60.5
Mask DINO [LZX+22]	-	59.5	60.8
FD-SwinV2-G [WHX+22]	896^2	-	61.4
BEIT-3	896^2	**62.0**	**62.8**

表 8：ADE20K 上语义分割的结果。“MS”是多尺度的缩写。对比系统的结果来自 paperswithcode.com 排行榜（时间戳：2022 年 08 月 22 日）。

Model	Extra Data	Image Size	ImageNet
*With extra **private** image-tag data*			
SwinV2-G [LHL+21]	IN-22K-ext-70M	640^2	90.2
ViT-G [ZKHB21]	JFT-3B	518^2	90.5
CoAtNet-7 [DLLT21]	JFT-3B	512^2	90.9
Model Soups [WIG+22]	JFT-3B	500^2	91.0
CoCa [YWV+22]	JFT-3B	576^2	91.0
*With only **public** image-tag data*			
BEIT [BDPW22]	IN-21K	512^2	88.6
CoAtNet-4 [DLLT21]	IN-21K	512^2	88.6
MaxViT [TTZ+22]	IN-21K	512^2	88.7
MViTv2 [LWF+22]	IN-21K	512^2	88.8
FD-CLIP [WHX+22]	IN-21K	336^2	89.0
BEIT-3	IN-21K	336^2	**89.6**

表 9：ImageNet-1K 上的 Top-1 准确率。

可能类别名称的文本嵌入与图像的特征嵌入。随后计算它们的余弦相似度分数，以预测每张图像最可能的标签。表 9 报告了在 ImageNet-1K 上的结果。我们首先在 ImageNet-21K 上进行中间微调，然后在 ImageNet-1K 上训练模型。为公平比较，我们仅与使用公共图像-标签数据的先前模型进行比较。BEIT-3 超越了以往模型，在仅使用公共图像-标签数据的情况下取得了新的最先进结果。

4 结论

在本文中，我们提出了 BEIT-3，这是一种通用多模态基础模型，在广泛的视觉和视觉-语言基准测试中实现了最先进的性能。BEIT-3 的核心思想是可以将图像建模为一种外语，从而能够以统一的方式对图像、文本以及图文对进行掩码“语言”建模。我们还证明，Multiway Transformers 能够有效建模不同的视觉和视觉-语言任务，使其成为通用建模的一个颇具吸引力的选择。BEIT-3 简单而有效，是扩展多模态基础模型的一个有前景的方向。对于未来工作，我们正在进行多语言 BEIT-3 的预训练，并将更多模态（例如音频）纳入 BEIT-3，以促进跨语言和跨模态迁移，并推动跨任务、语言和模态的大规模预训练的大融合。我们还希望通过结合 BEIT-3 和 MetaLM [HSD+22] 的优势，为多模态基础模型赋予上下文学习能力。

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A Effects of Intermediate Finetuning for Retrieval

As shown in Table 10, we directly finetune BEIT-3 on COCO and Flickr30K. BEIT-3 still outperforms previous state-of-the-art models, even without using image-text contrastive objective during pretraining. The results demonstrate the effectiveness of masked data modeling for learning cross-modal representations. Next, we perform intermediate finetuning on the pretraining image-text pairs for 5 epochs with a 16k batch size. The peak learning is 3e-5, with linear warmup over the first epoch. The image input size is 224×224. The weight decay is set to 0.05. We disable dropout as in pretraining and use drop path with a rate of 0.3. The layer-wise learning rate decay is 0.95. We use the AdamW [LH19] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$.

Model	MSCOCO (5K test set)						Flickr30K (1K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
BEIT-3	82.7	96.0	98.2	65.1	86.6	92.3	97.5	99.9	100.0	89.1	98.6	99.3
+ Intermediate Finetuning	**84.8**	**96.5**	**98.3**	**67.2**	**87.7**	**92.8**	**98.0**	**100.0**	**100.0**	**90.3**	**98.7**	**99.5**

Table 10: Finetuning results of image-text retrieval on COCO and Flickr30K. BEIT-3 is directly finetuned on downstream benchmarks without intermediate finetuning on the pretraining data.

B Hyperparameters Used for Pretraining

Hyperparameters	BEIT-3
Layers	40
Hidden size	1408
FFN inner hidden size	6144
Attention heads	16
Patch size	14×14
Relative positional embeddings	✗
Training steps	1M
Batch size	6144
AdamW ϵ	1e-6
AdamW β	(0.9, 0.98)
Peak learning rate	1e-3
Learning rate schedule	Cosine
Warmup steps	10k
Gradient clipping	3.0
Dropout	✗
Drop path	0.1
Weight decay	0.05
Data Augment	RandomResizeAndCrop
Input resolution	224^2
Color jitter	0.4

Table 11: Hyperparameters for pretraining BEIT-3.

A 中间微调对检索的影响

如表 10 所示，我们直接在 COCO 和 Flickr30K 上对 BEIT-3 进行微调。即使在预训练期间未使用图文对比目标，BEIT-3 仍然优于此前最先进的模型。结果证明了掩码数据建模在学习跨模态表示方面的有效性。接下来，我们在预训练图文对上进行 5 个 epoch 的中间微调，批量大小为 16k。峰值学习率为 3e-5，并在第一个 epoch 内进行线性预热。图像输入大小为 224 × 224。权重衰减设置为 0.05。与预训练一样，我们禁用 dropout，并使用比率为 0.3 的 drop path。逐层学习率衰减为 0.95。我们使用 AdamW [LH19] 优化器，其中 $\beta_1 = 0.9$，$\beta_2 = 0.999$。

Model	MSCOCO (5K test set)						Flickr30K (1K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
BEIT-3	82.7	96.0	98.2	65.1	86.6	92.3	97.5	99.9	100.0	89.1	98.6	99.3
+ Intermediate Finetuning	**84.8**	**96.5**	**98.3**	**67.2**	**87.7**	**92.8**	**98.0**	**100.0**	**100.0**	**90.3**	**98.7**	**99.5**

表 10：在 COCO 和 Flickr30K 上进行图文检索的微调结果。BEIT-3 直接在下游基准上进行微调，而未在预训练数据上进行中间微调。

B 用于预训练的超参数

Hyperparameters	BEIT-3
Layers	40
Hidden size	1408
FFN inner hidden size	6144
Attention heads	16
Patch size	14×14
Relative positional embeddings	✗
Training steps	1M
Batch size	6144
AdamW ϵ	1e-6
AdamW β	(0.9, 0.98)
Peak learning rate	1e-3
Learning rate schedule	Cosine
Warmup steps	10k
Gradient clipping	3.0
Dropout	✗
Drop path	0.1
Weight decay	0.05
Data Augment	RandomResizeAndCrop
Input resolution	224^2
Color jitter	0.4

表 11：预训练 BEIT-3 的超参数。

C Hyperparameters Used for Finetuning

Hyperparameters	NLVR2	VQAv2
Peak learning rate	1e-3	1e-5
Fine-tuning epochs	20	10
Warmup epochs	5	1
Layer-wise learning rate decay	0.8	1.0
Batch size	256	128
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.05	0.01
Drop path	0.4	
Dropout	✗	
Input resolution	224^2	756^2

Table 12: Hyperparameters for fine-tuning BEIT-3 on NLVR2 and VQAv2.

Hyperparameters	COCO Captioning
Peak learning rate	8e-6
Fine-tuning steps	16k
Warmup steps	1600
Layer-wise learning rate decay	1.0
Batch size	256
AdamW ϵ	1e-8
AdamW β	(0.9, 0.999)
Weight decay	0.01
Drop path	0.3
Dropout	✗
Input resolution	392^2
Mask prob	0.6
Label smoothing ε	0.1
Beam size	3

Table 13: Hyperparameters for fine-tuning BEIT-3 on COCO captioning.

Hyperparameters	COCO	Flickr30K
Peak learning rate	1e-5	
Fine-tuning epochs	15	20
Warmup epochs	3	5
Layer-wise learning rate decay	0.95	
Batch size	3k	
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.05	
Drop path	0.3	
Dropout	✗	
Input resolution	420^2	

Table 14: Hyperparameters for fine-tuning BEIT-3 on image-text retrieval.

C Hyperparameters Used for Finetuning

Hyperparameters	NLVR2	VQAv2
Peak learning rate	1e-3	1e-5
Fine-tuning epochs	20	10
Warmup epochs	5	1
Layer-wise learning rate decay	0.8	1.0
Batch size	256	128
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.05	0.01
Drop path	0.4	
Dropout	✗	
Input resolution	224^2	756^2

表 12：在 NLVR2 和 VQAv2 上微调 BEIT-3 的超参数。

Hyperparameters	COCO Captioning
Peak learning rate	8e-6
Fine-tuning steps	16k
Warmup steps	1600
Layer-wise learning rate decay	1.0
Batch size	256
AdamW ϵ	1e-8
AdamW β	(0.9, 0.999)
Weight decay	0.01
Drop path	0.3
Dropout	✗
Input resolution	392^2
Mask prob	0.6
Label smoothing ε	0.1
Beam size	3

表 13：在 COCO 图像描述任务上微调 BEIT-3 的超参数。

Hyperparameters	COCO	Flickr30K
Peak learning rate	1e-5	
Fine-tuning epochs	15	20
Warmup epochs	3	5
Layer-wise learning rate decay	0.95	
Batch size	3k	
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.05	
Drop path	0.3	
Dropout	✗	
Input resolution	420^2	

表 14：在图文检索上微调 BEIT-3 的超参数。

Hyperparameters	ADE20K
Peak learning rate	1e-5
Fine-tuning steps	80k
Warmup steps	1500
Layer-wise learning rate decay	0.95
Batch size	16
AdamW ϵ	1e-8
AdamW β	(0.9, 0.999)
Weight decay	0.05
Drop path	0.5
Dropout	✗
Input resolution	896^2

Table 15: Hyperparameters for fine-tuning BEIT-3 on semantic segmentation.

Hyperparameters	Object365	COCO
Learning rate	1e-4	5e-5
Fine-tuning epochs	15	20
Warmup steps	250	
Layer-wise learning rate decay	0.9	
Batch size	64	
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.1	
Drop path	0.6	
Input resolution	1024^2	1280^2

Table 16: Hyperparameters for fine-tuning BEIT-3 on object detection.

Hyperparameters	ImageNet-21K	ImageNet-1K
Peak learning rate	5e-5	3e-5
Fine-tuning epochs	50	15
Warmup epochs	5	3
Layer-wise learning rate decay	0.85	0.95
Batch size	16k	2k
AdamW ϵ	1e-6	1e-8
AdamW β	(0.9, 0.98)	(0.9, 0.999)
Weight decay	0.05	
Drop path	0.4	
Dropout	✗	
Input resolution	224^2	336^2
Label smoothing ε	0.1	

Table 17: Hyperparameters for fine-tuning BEIT-3 on image classification.

Hyperparameters	**ADE20K**
Peak learning rate	1e-5
Fine-tuning steps	80k
Warmup steps	1500
Layer-wise learning rate decay	0.95
Batch size	16
AdamW ϵ	1e-8
AdamW β	(0.9, 0.999)
Weight decay	0.05
Drop path	0.5
Dropout	✗
Input resolution	896^2

表15：在语义分割上微调 BEIT-3 的超参数。

Hyperparameters	**Object365**	**COCO**
Learning rate	1e-4	5e-5
Fine-tuning epochs	15	20
Warmup steps	250	
Layer-wise learning rate decay	0.9	
Batch size	64	
AdamW ϵ	1e-8	
AdamW β	(0.9, 0.999)	
Weight decay	0.1	
Drop path	0.6	
Input resolution	1024^2	1280^2

表 16：在目标检测任务上微调 BEIT-3 的超参数。

Hyperparameters	**ImageNet-21K**	**ImageNet-1K**
Peak learning rate	5e-5	3e-5
Fine-tuning epochs	50	15
Warmup epochs	5	3
Layer-wise learning rate decay	0.85	0.95
Batch size	16k	2k
AdamW ϵ	1e-6	1e-8
AdamW β	(0.9, 0.98)	(0.9, 0.999)
Weight decay	0.05	
Drop path	0.4	
Dropout	✗	
Input resolution	224^2	336^2
Label smoothing ε	0.1	

表 17：在图像分类上微调 BEIT-3 的超参数。